Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



# Team Details

**Team name:** ReKindle

**Team leader:** Abhishek Puvvada

**Problem statement:** AI-powered customer re-engagement and segmentation for customer-centric commerce

# Brief about the idea

**ReKindle turns a merchant's past-customer database into recurring revenue.** It finds customers who have gone quiet, writes a personal WhatsApp message for each (email optional), and sends only what has been fact-checked. Human-supervised first; autonomy is earned.

## 1 Find

BigQuery SQL scores recency, frequency and spend from Baby Trails bookings and spots the next natural session, like a newborn reaching baby-session age. No LLM.

## 2 Write

Gemini on Vertex AI drafts the message. Every claim must cite a fact from that customer's own record. It never invents a name, price or memory.

## 3 Verify

A deterministic guardrail blocks anything ungrounded. Only customers who opted in are ever drafted for, and STOP is honoured before any generation.

## 4 Earn trust

Nothing auto-sends until the owner has reviewed 20 drafts. Then only low-risk drafts do. Milestones and discounts always go to the owner.

Built on Google Cloud: BigQuery, Gemini on Vertex AI, Cloud Run, Firestore, Looker Studio. POC built with Babytrail Photography (Baby Trails), whose booking system already runs on GCP; a pilot is planned. A studio is the hardest test of personalization because every message touches a family milestone; the same engine generalizes to apparel, D2C, salons and clinics.

# Opportunities

| | |
|---|---|
| **How is it different?** | Re-engagement tools blast templates. Most AI entries are chat wrappers or static dashboards. ReKindle is a hybrid agent: deterministic logic decides who, when and what is allowed; Gemini only writes; a deterministic verifier and the owner decide what is sent. |
| **How does it solve the problem?** | Past customers are a merchant's cheapest revenue but nobody has time to write 300 personal messages. ReKindle segments the base, drafts each message grounded in that customer's history, and routes it: low-risk drafts go out automatically once trust is earned, everything else lands in a one-tap cockpit. |
| **What is the USP?** | AI speed without the hallucination risk that keeps merchants from letting AI near their customers. In our adversarial test, 1,864 corrupted drafts were all stopped before sending, and 0 reached a customer without a human. |

# List of features offered by the solution

## Automated segmentation

RFM scoring and dormancy triggers in one BigQuery query; journey triggers from the baby's estimated age

## Grounded drafts with citations

Gemini structured output: every claim quotes the message and cites a fact id

## Fact-verification guardrail

Blocks invented numbers, names, offers, links, pressure and fabricated memories

## Risk-tiered routing

Low / medium / high. Milestones and discounts always need the owner

## Earned autonomy

Auto-send unlocks after 20 owner decisions; threshold learns from approvals, edits and rejections

## Opt-in first

No opt-in, no draft: the photo-use flag is never treated as messaging consent. STOP suppresses before generation and cancels queued drafts

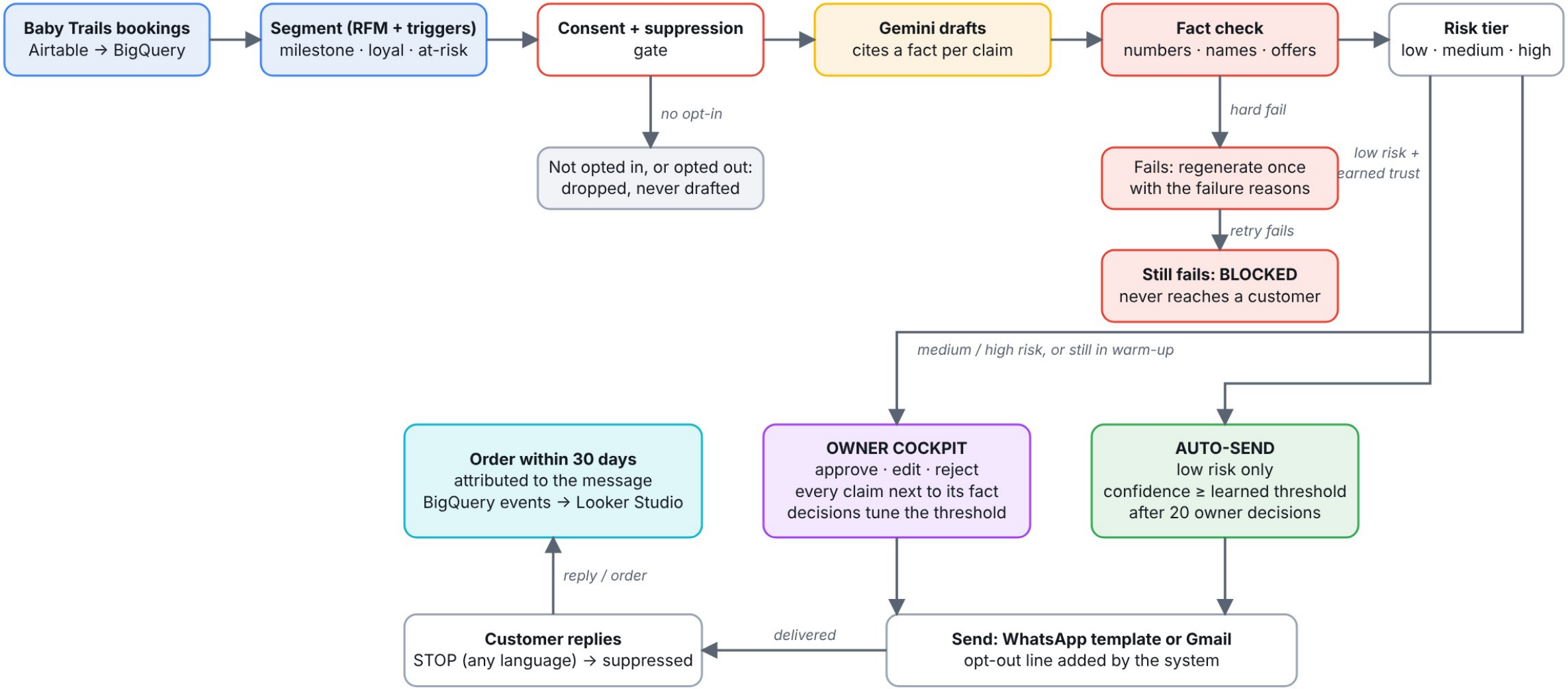
## WhatsApp-first delivery

Approved Marketing template for dormant customers (24h rule); email optional; opt-out line added by the system

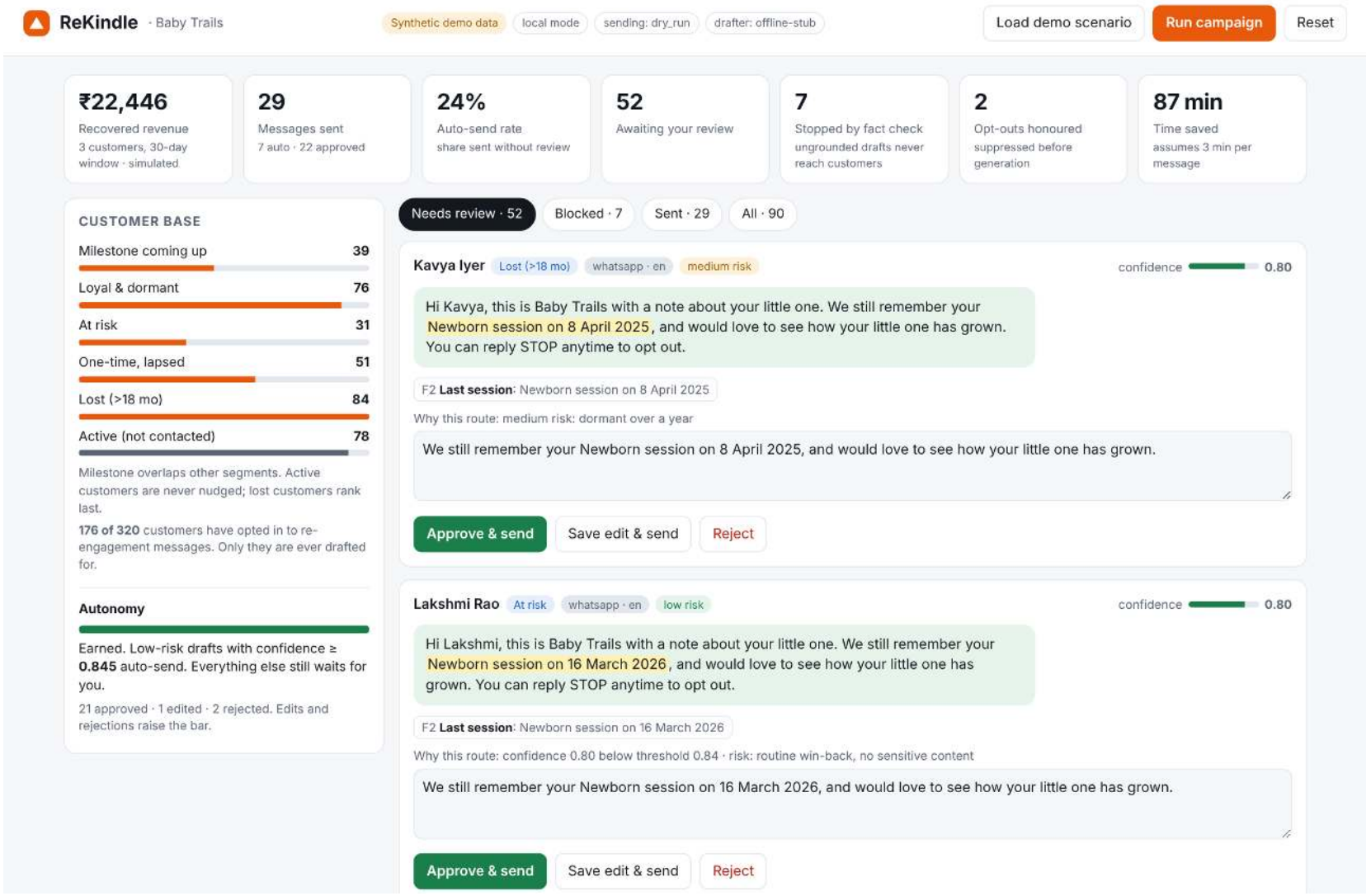
## Owner cockpit + dashboard

Real-time Firestore queue with claims shown next to facts; BigQuery view for Looker Studio

# Process flow diagram



# Wireframes: the owner cockpit (live prototype)




**1 Highlighted words:** each claim in the message, hover to see its source fact

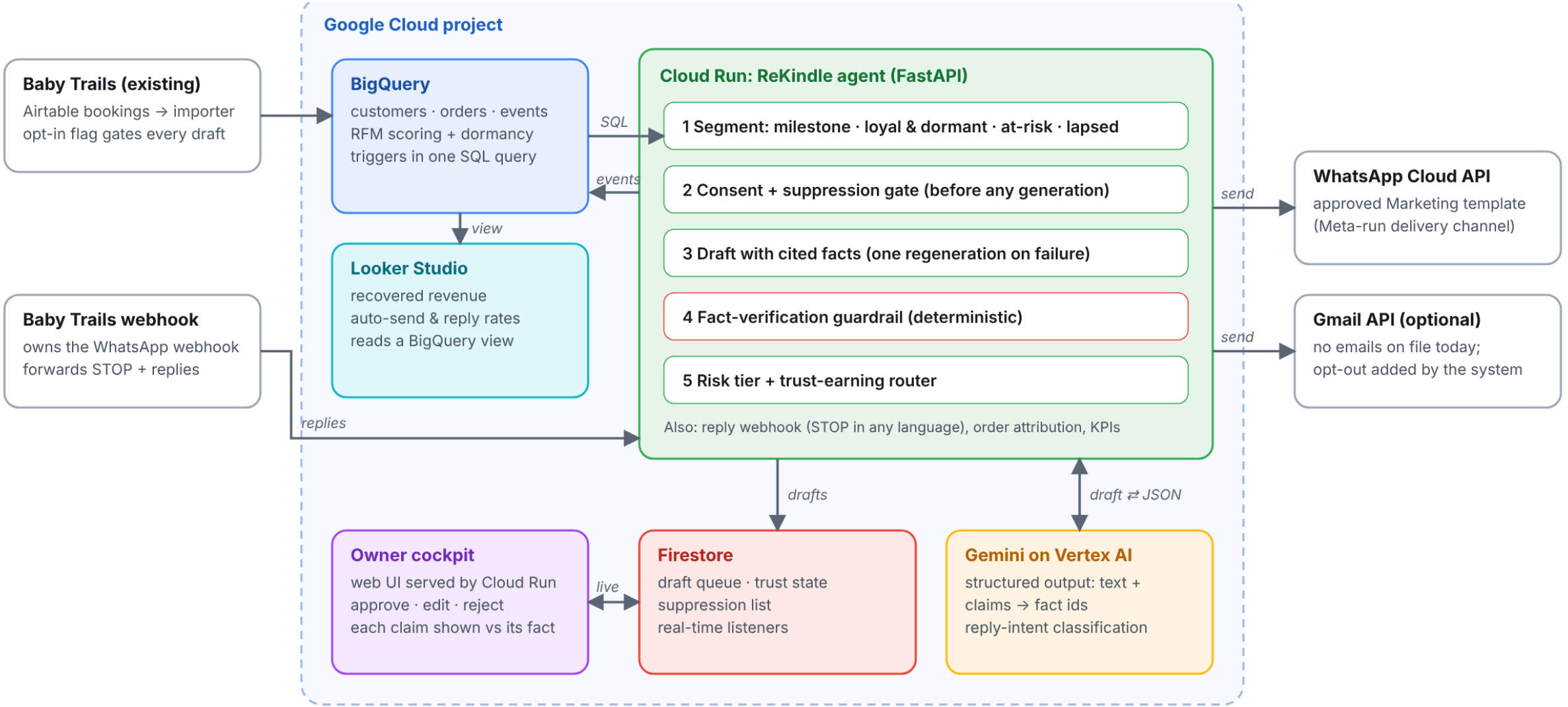
**2 Fact chips:** the customer-record fields the draft is allowed to use

**3 Risk + confidence:** why this draft needs a human

**4 Autonomy meter:** warm-up progress, then the learned auto-send threshold

**5 Approve / Edit / Reject:** edits are re-verified; every decision tunes the threshold

# Architecture diagram of the proposed solution



# Built on a real studio: the Baby Trails integration

## What we read

- Baby Trails' Airtable Bookings: phone, shoot type, baby age, package, totals, paid flags
- Its catalog: newborn 5-30 days, baby 3-12 months, maternity
- Same WhatsApp number and GCP project; ReKindle runs beside it

## What we never assume

- Its photo-use flag is not messaging consent
- No child name on file, so drafts say 'your little one'
- Only an age at booking, so milestones are estimates, quoted by month
- Maternity history never produces a baby message

## What the pilot needs

- An opt-in question at booking: today none of the studio's customers are eligible
- Inbound routing: STOP and replies forwarded to ReKindle
- An approved Marketing template (rk_reengage_v1)
- Venkat's real prices and approved offers

Reviewed against the Baby Trails repo. Details and proposed changes: docs/BABYTRAILS_INTEGRATION.md

# Technologies used in the solution

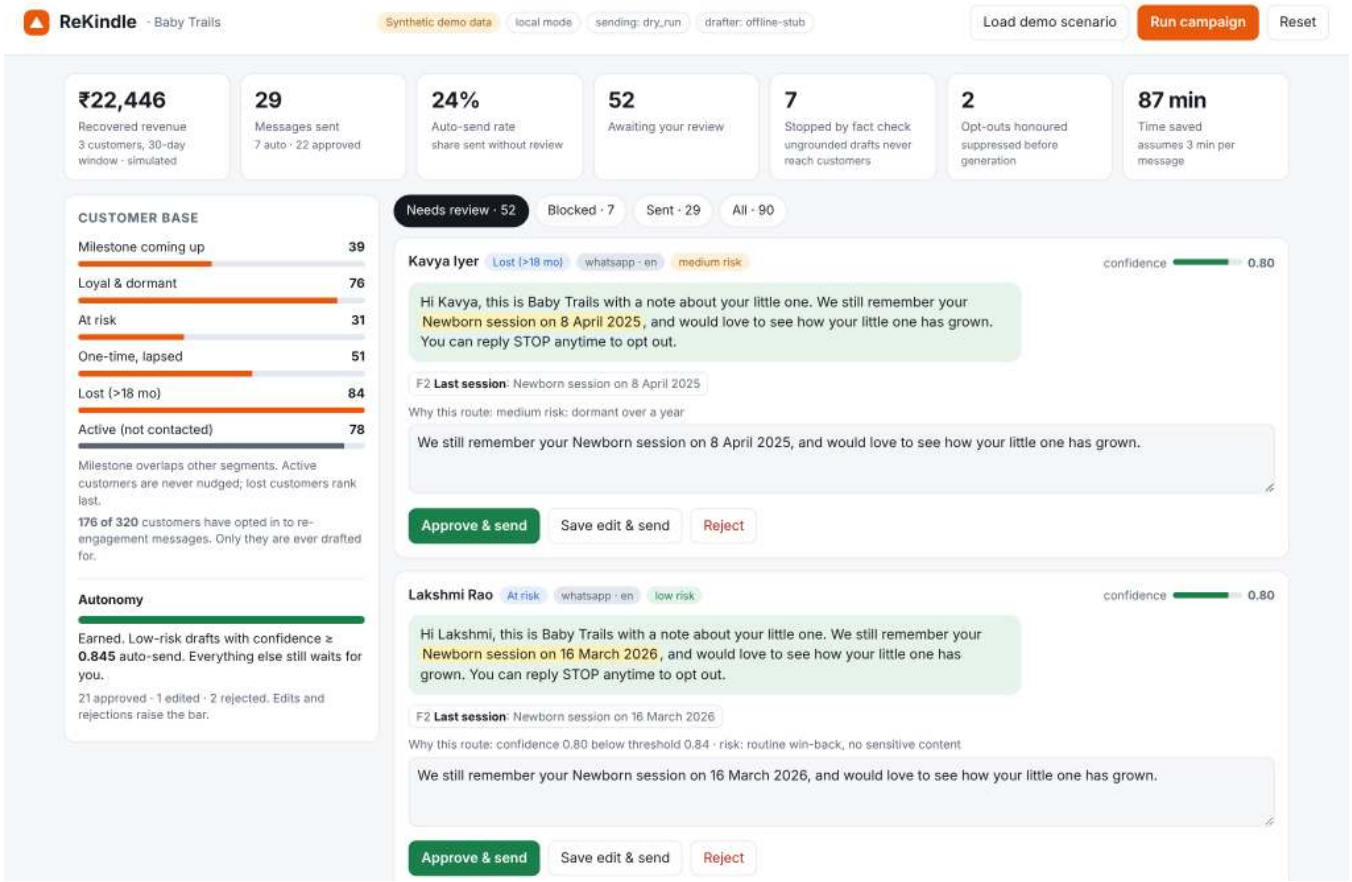
| Technology | Role in ReKindle |
|---|---|
| **Gemini 3.8 Flash on Vertex AI** | Message drafting with structured output (response_schema) that forces claim-to-fact citations; reply-intent classification |
| **BigQuery** | Source of truth for customers and orders; deterministic RFM + dormancy SQL; append-only event log |
| **Cloud Run** | Agent orchestration (FastAPI), owner cockpit UI, WhatsApp webhook |
| **Firestore** | Real-time draft queue, trust state and suppression list for the cockpit |
| **Looker Studio** | Dashboard over a BigQuery view: recovered revenue, auto-send and reply rates |
| **Baby Trails (existing)** | Airtable bookings are the customer and order source; its server owns the WhatsApp number and webhook |
| **Gmail API (optional)** | Email with the system-added opt-out line; Baby Trails holds no emails today |
| **WhatsApp Cloud API** | Delivery channel only (Meta-run), approved templates; the rest of the stack is Google |

# Estimated implementation cost

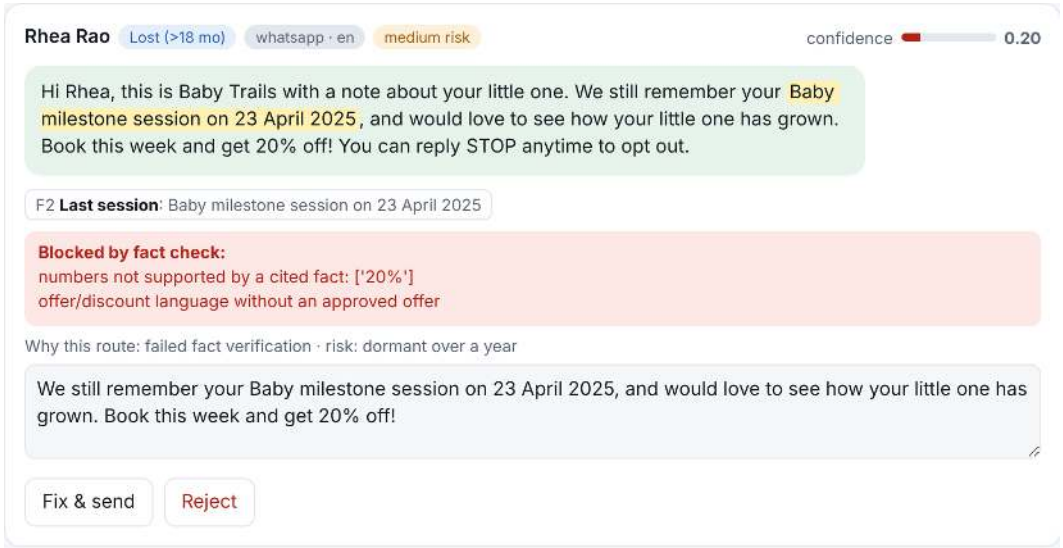
The MVP runs within Google Cloud hackathon credits. Every cost driver is consumption-based, and the design keeps the expensive part (the LLM) to one call per message, two at most.

- **Gemini on Vertex AI:** scales with messages drafted; one call per message, plus one regeneration only when the guardrail blocks a draft
- **BigQuery:** scales with customer records scanned; segmentation is one query per campaign run
- **Cloud Run:** pay-per-request, scales to zero between campaigns
- **Firestore:** scales with cockpit activity (drafts written, decisions made)
- **WhatsApp Business API:** per-conversation fees from the provider; email via Gmail API has no per-message fee

# Snapshots of the prototype




Owner review queue: claims highlighted against their source facts; autonomy meter shows trust earned




Guardrail at work: an invented 20% discount is blocked and never reaches the customer

Running prototype on synthetic customer data, not real customers.

# Prototype performance report / benchmarking

| 1,864 | 1,864 | 0 | 0% |
|---|---|---|---|
| adversarial drafts tested | stopped by the guardrail | reached a customer without a human | good drafts wrongly blocked |

- 13 corruption types a hallucinating model could produce (invented discount, wrong age, invented name, fake product, pressure, links, guarantees, fabricated memories) applied to all 145 outreach targets, with auto-send fully unlocked and the threshold at its floor.
- Seeded demo run on synthetic Baby Trails-shaped data (320 customers, 176 opted in): 90 drafts, 29 sent (7 auto-sent, 22 owner-approved), 7 blocked, 2 opt-outs honoured. Owner decisions, replies and orders are simulated; the opt-in rate is illustrative.
- 70 automated tests cover the Baby Trails adapter, segmentation, opt-in and STOP, the guardrail, routing and the approval API.

**Honest scope.** This tests the guardrail with our own corruption set on synthetic data, not Gemini's real output or real customers. Pilot metrics (auto-send rate, approval latency, time saved, recovered revenue) are planned for the pilot with Baby Trails and will replace the simulated figures.

# Additional details / future development

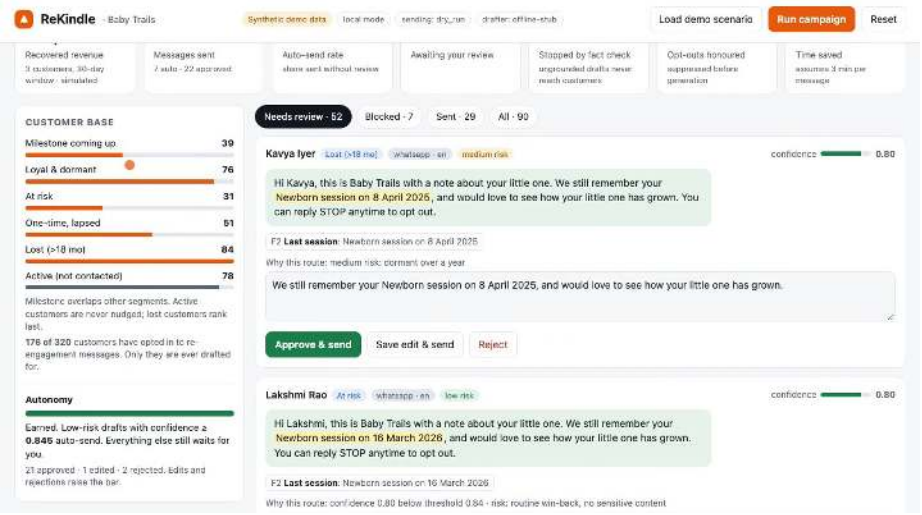
- **Second-layer semantic verifier:** a Gemini entailment check on top of the deterministic one, for paraphrased claims no rule can see
- **Native integrations:** Shopify, WooCommerce and Unicommerce for one-click onboarding
- **Context-aware retrieval:** Vertex AI Search over catalog and order history for richer personalization
- **Inventory-aware campaigns:** promote what is in stock, pause what is not
- **More channels:** SMS and Instagram DMs alongside WhatsApp and email
- **Predictive intelligence:** churn and lifetime-value scoring with BigQuery ML beyond RFM
- **Trust-earning autonomy at scale:** merchant edits teach the model; the cockpit becomes an exception-handling tool, not a daily chore

# Provide links to your

**GitHub Public Repository:** https://github.com/JayMandava/rekindle

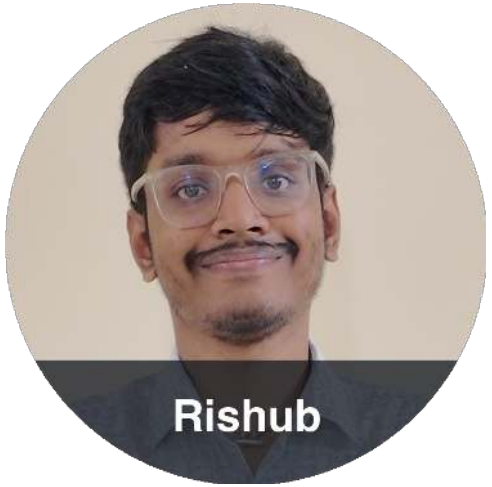
**Demo Video Link (under 90 seconds):**
https://github.com/JayMandava/rekindle/blob/main/docs/demo/rekindle_demo.mp4

**Final Product Link:** https://jmandava-rekindle.hf.space



# Team









Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.

# Thank you

